Civil Engineering x Hazard Modeling

Tephra2 on VICTOR

Objectives:

1. **Interact** with an advanced hazard forecast model & **identify** which inputs are necessary to generate meaningful outputs of such models.

2. **Identify** how parameter inversions for magma chamber models can be used to calibrate eruption models and evaluate possible eruption Scale and societal impact.

3. **Practice** and **apply** how engineers can use earth process models to inform infrastructure development.

Introduction and Background:

In this lab, you will use a numerical model called 'Tephra2", which simulates the dispersion of volcanic ash from an eruption, to assess how eruption conditions impact the built infrastructure in surrounding communities. The lab uses a hypothetical eruption at Craters of the Moon vent as the test case.



source: USGS

Preparation and Materails:

you will need...

- Access to the Internet.
- To be registered on VICTOR.
  - If you have not completed this step yet, navigate to 'victor.ldeo.columbia.edu' and fill out the form under 'sign up' in the menu.
  - ✱ The VICTOR team needs to manually confirm all registrations, so this may take up to 24 hours.
- To be signed up for the VICTOR forum.
  - forum.victorproject.org/signup
  - Use this as a resource to ask for help or search for previously asked questions from other users.
- Optional tasks:
  - Read or review the Tephra2 Manual: tinyurl.com/y49a83cy
  - Watch the "Tephra2 on VICTOR' tutorial on Youtube.
    - tinyurl.com/2zx8psdv

Tephra2 Users Manual

Tephra2 on VICTOR

Tasks and Procedures

Part One:

Picking location points:

Site Name	Town	Description	Lat.	Long.	Roof Area
Site #1	MacKay				
Site #2	Acro				

- Use the table above to record target locations.

- Using Google Maps, identify one structure in each town (**MacKay and Acro, Idaho**).
- Record the **coordinates** the table (right-click on the structure, lat and long are shown at the top of the drop-down menu).
- Describe the building's use under **description**: for example school, government building, airport etc.
- In google maps, right click on the structure. Use measure distances to find the roof area by outlining the perimiter of the building's roof. The **area** will be shown in a small widow at the bottom of your Google Maps screen.

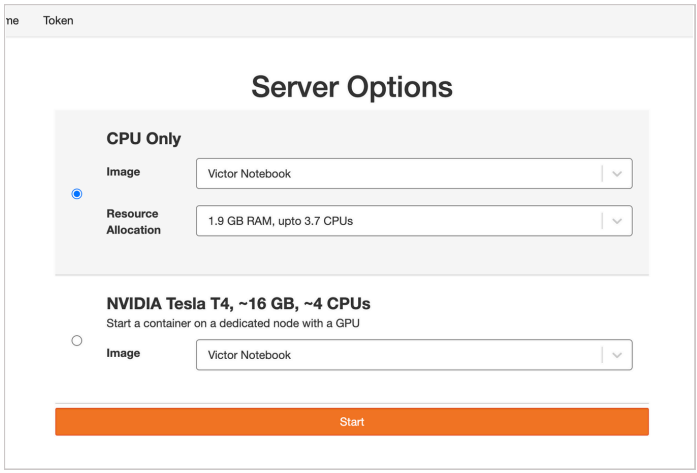


Tasks and Procedures

Part Two:

Getting Started on VICTOR:

1. Log in to VICTOR at hub.victorproject.org
2. **"Server Options"**
   - Ensure Victor Notebook is selected
   - Under resource allocation, select



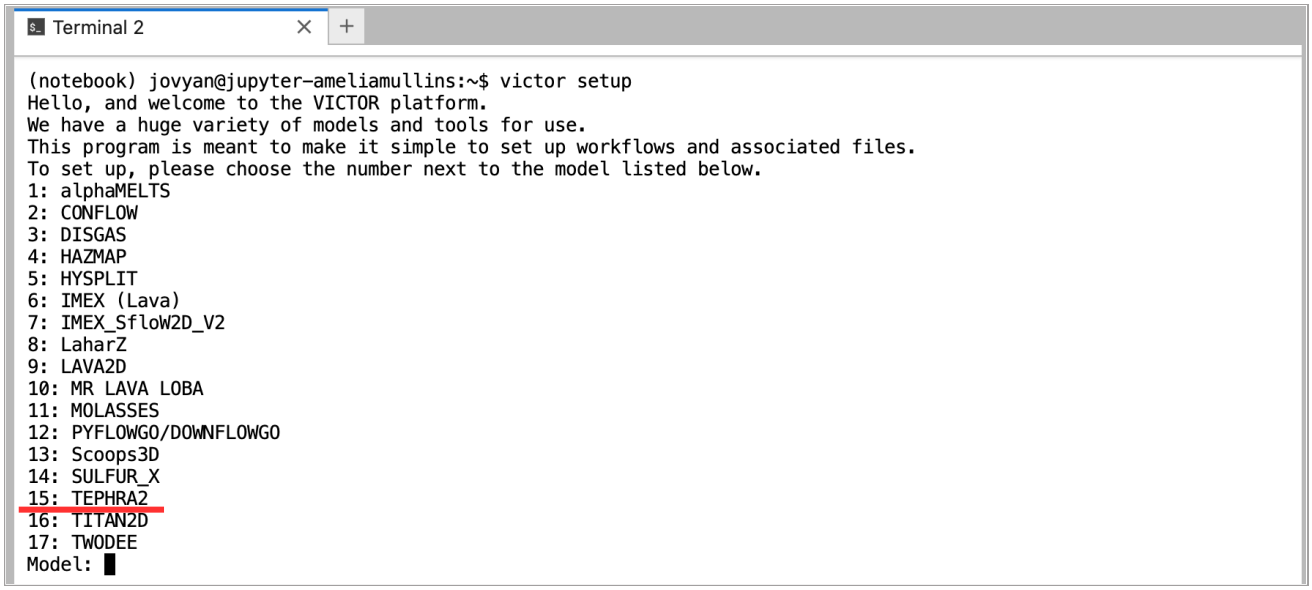
7.4 GB RAM, upto 3.7 CPUs

   - Clicking Start will bring you to The Launcher.

3. The Launcher and VICTOR Setup command



- From the Launcher, open a new terminal by clicking the **terminal icon** at the bottom left of your launcher.
- In the new terminal, type. `victor setup` and press enter.
- Identify TEPHRA2 in the list that appears
- Type its corresponding number and press enter.



- On the left of your screen, a folder called tephra2 will appear.
- Open the folder by clicking it and find the Jupyter Notebook for tephra2.ipynb
- Before continuing, read the information at the top of the notebook.

Tasks and Procedures

Part Three:

Setting Up and Running the Model

First: Change the vent location to Craters of the Moon: `= 43.31,-113.47`

Next: For your data, we will explore eruption volumes in the range of **1 km^3, 3km^3 and 5km^3**, however, Tephra2 takes eruption mass as input. Convert these values and record them in the table provided (Results and Questions: Page 5).

Follow these steps to run the model with each value of erupted mass.
a. Set the mass eruption value to the target variable, and run the configuration file code cells.

b. Obtain a wind file by setting the date to 10/23/2024 using the relevant variables, and run the wind data code calls.

```
[2]: #INPUT DATE AND COORDINATE VALUES HERE

#Coordinate location
vent_latitude, vent_longitude = 43.31,-113.47

#Specify month(s) (in numerical string format, e.g. "01", "11")
months = ["10"]

#Specify year(s) (in numerical string format, e.g. "1999")
years = ["2024"]

#Specify day(s) (in numerical string format, e.g. "01", "11")
days = ["23"]

#Specify hour(s) in 24 hour string format (00:00 - 23:00)
hours = ['10:00']
```

c. Set up target of interest. Enter the coordinates of your three sites from the table in part 1. (The image below uses **example coordinates,** and only demonstrates correct formatting.)

```
locations = [[44.05,-121.32],[44.27,-121.17],[43.87,-121.43]]

elevation = 1000

locations = np.array(locations)
loced = utm.from_latlon(locations[:,0],locations[:,1])
```

d. Run Tephra2. (<one minute).
(if the cell number appears as an asterisk, the cell is still running)
e. Use the table on page 5 to record the output valves for each site.

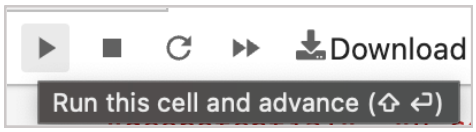
```
[*]:
```

Tasks and Procedures

Part Three:

Notes for Running the Notebook:

- Run each cell by clicking the play button at the top of the notebook or press Shift + Enter at the same time.



- Wait until you see a sequence number appear in the brackets, if the cell number is replaced with an asterisk, the cell is still running!



[7]:
```
if a cell finishes running and you see a red
message beneath it:
```

```
review the message within the red, if it does not read 'error' within the first
few lines, it is considered a warning, and the notebook can continue running!

If you receive an unexpected error message:
Consult the VICTOR forum
or
Contact the VICTOR team at victor@ldeo.columbia.edu - include screenshots in your
email!
```

Data and Questions

Use this table to record the data collected in part three, and your calcualtions from page 6.

Site Name	Erupted Volume	Erupted Mass	Deposit Mass/area (kg/m2)	Deposit load (kPa)	Total mass on the roof (kg)
Site 1	**1km^3**				
	3km^3				
	5km^3				
Site 2	**1km^3**				
	3km^3				
	5km^3				

Useful Information:

Gravitational Acceleration:

9.8 m/s ^{2}

Density of Tephra:

1200 kg/m ^{3}

Erupted mass:

(density of tephra) x (erupted volume)

Deposit load:

Deposit mass x gravity

Data and Questions

Answer The Following:

Using the iso mass map from the parsed tephra2 output, answer the following questions:

1. Assess: Which direction did most of the output travel from for each location? (Your answer will be a cardinal direction.)

Site 1 (MacKay):

Site 2 (Acro):

2. From the information given in the map, calculate mass per unit area of deposit at each site (and record in table)
3. Calculate the load (weight per area) of the deposit on the roof by multiplying by the gravitational acceleration. (Record in table).
4. From the mass/area value and the roof dimensions you wrote down, calculate the total ash mass weighing the roof at each site (and record in table)